

Bank Loan Analysis — End-to-End Data Analyst Project

1. Project Overview

This project analyzes a bank loan portfolio using Python and focuses on a complete, business-oriented Data Analyst workflow.

The analysis moves from:

Business Understanding → Data Inspection → Data Quality → Data Cleaning → Feature Engineering → Core Portfolio Analysis → Additional Risk Analysis → Supporting EDA → KPI Summary → Key Insights → Business Interpretation → Recommendations

The project is designed to answer business questions using simple, explainable analysis rather than machine learning or unnecessarily complex statistical methods.

2. Business Problem

A bank wants to understand its loan portfolio from four main perspectives:

- **Portfolio size and financial exposure**
- **Loan origination activity and portfolio composition**
- **Observed Good vs Bad Loan performance**
- **Risk and borrower patterns across important segments**

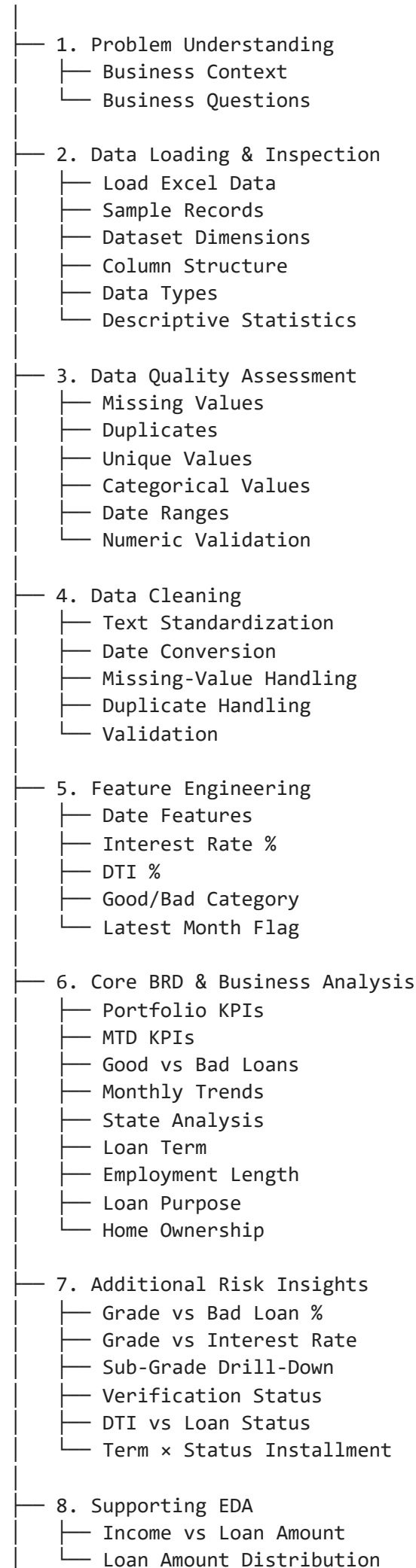
The analysis should help the business understand **where lending activity is concentrated, how observed loan outcomes differ across segments, and which patterns should be monitored further.**

Main Business Questions

Area	Business Question
Portfolio	How large is the loan portfolio?
Financial	How much was funded and how much loan-level payment is recorded?
Pricing	What are the average interest rate and DTI?
Performance	What proportion is classified as Good vs Bad?
Trends	How has loan origination changed over time?
Geography	Which states have the largest activity and what is the observed risk?
Term	How do 36-month and 60-month loans differ?
Employment	How does lending vary by employment length?
Purpose	Which loan purposes dominate activity and what are their observed risk rates?
Housing	How does activity and observed risk vary by home ownership?
Risk	How do Grade, Sub-Grade, Verification Status and DTI relate to observed outcomes?

3. Project Mind Map

BANK LOAN ANALYSIS



- 9. KPI Summary
- 10. Key Insights
- 11. Business Interpretation
- 12. Business Recommendations
- 13. Conclusion

2. Data Loading & Inspection

Dataset Profile

Item	Result
Rows	38,576
Original columns	24
Source	Excel loan-level dataset
Grain	One row represents one loan/application record
Loan ID uniqueness	Unique
Member ID uniqueness	Unique

Key fields used

Field	Business meaning in analysis
id	Loan/application identifier
address_state	Borrower state
emp_length	Employment length category
grade	Loan grade
sub_grade	More detailed grade segment
home_ownership	Housing category
issue_date	Loan origination date
loan_status	Original loan outcome/status
purpose	Stated loan purpose
term	Repayment term
verification_status	Verification category
annual_income	Borrower annual income
dti	Debt-to-income ratio
installment	Loan installment amount
int_rate	Interest rate

Field	Business meaning in analysis
loan_amount	Funded loan principal
total_payment	Cumulative loan-level payment amount

Initial data observations

- All 38,576 records were loaded successfully.
- `emp_title` is the only field with missing values.
- Main categorical dimensions have clean, interpretable categories.
- `issue_date` covers **01-Jan-2021 to 12-Dec-2021**.
- The latest observed issue date is **12-Dec-2021**.

3. Data Quality Assessment

5.1 Missing Values

Field	Missing	Percentage	Decision
<code>emp_title</code>	1,438	3.73%	Retained
All other fields	0	0%	No missing-value action needed

Missing-value approach

- Loan records were **not dropped** because employer title is not required for the core portfolio analysis.
- No unreliable imputation was performed.
- A separate reporting field, `emp_title_analysis`, uses **Not Available** when employer title is needed for analysis.
- The original `emp_title` field is preserved.

5.2 Duplicate Assessment

Check	Result
Duplicate rows	0
Duplicate Loan IDs	0
Duplicate Member IDs	0

Decision: no records were removed for duplication.

5.3 Key Categorical Structure

Dimension	Number of categories
Application Type	1
Loan Term	2

Dimension	Number of categories
Loan Status	3
Verification Status	3
Home Ownership	5
Grade	7
Employment Length	11
Purpose	14
Sub-Grade	35
State	50

Important category distributions

- Loan Status: **32,145 Fully Paid, 5,333 Charged Off, 1,098 Current.**
- Term: **28,237 36-month, 10,339 60-month.**
- Grade: B has the largest volume with **11,674** applications.
- State: California has the largest activity with **6,894** applications.
- Purpose: Debt consolidation dominates with **18,214** applications.
- Home Ownership: RENT dominates with **18,439** applications.

5.4 Numeric Validation

Validation Rule	Result
loan_amount <= 0	0
Negative int_rate	0
Negative dti	0
Missing issue_date	0

The numeric checks were diagnostic; no valid observations were removed based on these rules.

4. Data Cleaning

The cleaning approach was intentionally conservative.

Cleaning performed

- Trimmed leading/trailing whitespace from important text fields.
- Converted date fields to `datetime`.
- Confirmed no missing dates were introduced during conversion.
- Retained missing `emp_title` records.
- Created `emp_title_analysis` only for reporting purposes.
- Confirmed duplicate rows and identifiers remained at zero.
- Rechecked cleaned fields after transformation.

Final post-cleaning state

Check	Result
Rows	38,576
Columns	25 after creating the reporting/analytical field
Duplicate rows	0
Duplicate Loan IDs	0
Missing <code>emp_title</code>	1,438
Missing <code>issue_date</code>	0

5. Feature Engineering

The project creates business-friendly fields while preserving the original variables.

Created Feature	Purpose
<code>issue_year</code>	Year-level analysis
<code>issue_month</code>	Month-level analysis
<code>issue_month_name</code>	Presentation-friendly month
<code>issue_year_month</code>	Monthly grouping
<code>interest_rate_pct</code>	Interest rate displayed as percentage
<code>dti_pct</code>	DTI displayed as percentage
<code>loan_category</code>	Good/Bad analytical classification
<code>is_latest_issue_month</code>	Latest observed month filtering

Good / Bad analytical rule

Good Loan = Fully Paid + Current
Bad Loan = Charged Off

This is an **analytical project rule** because the supplied BRD requests Good/Bad KPIs but does not explicitly define the status mapping.

Reporting period

- Latest issue date: **12-Dec-2021**
- Latest available calendar month: **December 2021**
- December is therefore a **partial observed period**.

6. Core BRD & Business Analysis

8.1 Executive Portfolio KPIs

KPI	Value
Total Loan Applications	38,576
Total Funded Amount	\$435.76M
Total Amount Received*	\$473.07M
Average Interest Rate	12.05%
Average DTI	13.33%

Insight

The observed portfolio contains **38,576 loan applications** with approximately **\$435.76M** in funded principal. Average interest rate is **12.05%**, while average DTI is **13.33%**.

Payment note: `total_payment` is a cumulative loan-level field, not a transaction-level cash ledger. The \$473.07M figure should therefore be described as recorded cumulative loan-level payment.

8.2 MTD KPI Analysis

MTD Metric	Value
Reporting Date	12-Dec-2021
MTD Loan Applications	4,314
MTD Funded Amount	\$53.98M
MTD Total Amount Received*	\$58.07M

Insight

The latest observed month contains **4,314 applications** and approximately **\$53.98M** in funded principal.

Because the dataset ends on **12-Dec-2021**, December MTD figures are **partial-month figures**.

8.3 Good vs Bad Loan KPIs

Metric	Good Loan	Bad Loan
Applications	33,243	5,333
Portfolio Share	86.18%	13.82%
Funded Amount	\$370.22M	\$65.53M
Loan-Level Payment	\$435.79M	\$37.28M

Insight

Under the project's analytical mapping:

- **86.18%** of loans are classified as Good Loans.

- **13.82%** are classified as Bad Loans.
 - Bad Loans represent approximately **\$65.53M** of funded principal.
 - Bad Loans represent a somewhat larger share of funded principal than their share of total loan count, so volume and dollar exposure should be considered together.
-

8.4 Monthly Trends by Issue Date

Observed monthly movement

Month	Applications	Funded Amount
Jan-2021	2,332	\$25.03M
Feb-2021	2,279	\$24.65M
Mar-2021	2,627	\$28.88M
Apr-2021	2,755	\$29.80M
May-2021	2,911	\$31.74M
Jun-2021	3,184	\$34.16M
Jul-2021	3,366	\$35.81M
Aug-2021	3,441	\$38.15M
Sep-2021	3,536	\$40.91M
Oct-2021	3,796	\$44.89M
Nov-2021	4,035	\$47.75M
Dec-2021*	4,314	\$53.98M

Insight

- Origination activity generally increased across the observed 2021 period.
 - Applications rose from **2,332 in January** to **4,314 in the observed December period**.
 - Funded amount rose from approximately **\$25.03M** to **\$53.98M**.
 - December is partial through **12-Dec-2021**, so it should not be treated as a full-month peak.
-

8.5 State Analysis

Highest-volume states

State	Applications	Funded Amount
CA	6,894	\$78.48M
NY	3,701	\$42.08M
FL	2,773	\$30.05M
TX	2,664	\$31.24M
NJ	1,822	\$21.66M
IL	1,486	\$17.12M

State	Applications	Funded Amount
PA	1,482	\$15.83M
VA	1,375	\$15.98M
GA	1,355	\$15.48M
MA	1,310	\$15.05M

State risk screening

The screening rule flags states with:

- at least **2% of total portfolio applications**, and
- an observed Bad Loan rate **above the overall portfolio rate of 13.82%**.

State	Applications	Funded Amount	Bad Loan %
FL	2,773	\$30.05M	17.27%
CA	6,894	\$78.48M	15.33%
GA	1,355	\$15.48M	15.28%
MD	1,027	\$11.91M	15.09%
NJ	1,822	\$21.66M	14.98%
WA	805	\$8.85M	14.66%

Insight

California is the largest portfolio by application volume. State risk should be interpreted together with **application volume and funded exposure**, not from percentages alone.

8.6 Loan Term Analysis

Term	Applications	Funded Amount	Bad Loan %	Avg Interest Rate	Avg DTI
36 months	28,237	\$273.04M	10.71%	11.03%	12.99%
60 months	10,339	\$162.72M	22.34%	14.83%	14.26%

Installment view

Term	Status	Applications	Avg Installment	Median Installment
36 months	Charged Off	3,023	\$316.79	\$263.56
36 months	Fully Paid	25,214	\$313.95	\$264.43
60 months	Charged Off	2,310	\$372.17	\$348.44
60 months	Current	1,098	\$398.92	\$369.96
60 months	Fully Paid	6,931	\$351.72	\$318.64

Insight

- 36-month loans account for the majority of applications.
- 60-month loans show higher average interest rate, higher observed Bad Loan %, and higher average DTI.
- Installments are also generally higher for the 60-month groups in this dataset.
- These measures should be considered together rather than using term volume alone.

8.7 Employment Length Analysis

Largest employment segments

Employment Length	Applications	Funded Amount	Bad Loan %
10+ years	8,870	\$116.11M	14.90%
< 1 year	4,575	\$44.21M	13.79%
2 years	4,382	\$44.97M	12.80%
3 years	4,088	\$43.94M	13.41%
4 years	3,428	\$37.60M	13.24%
5 years	3,273	\$36.97M	13.72%
1 year	3,229	\$32.88M	13.78%
6 years	2,228	\$25.61M	13.73%
7 years	1,772	\$20.81M	14.79%
8 years	1,476	\$17.56M	13.55%
9 years	1,255	\$15.08M	12.35%

Insight

The **10+ years** group has the largest application volume and funded exposure. Employment length provides useful segmentation, while risk percentages should be considered alongside group size.

8.8 Loan Purpose Analysis

Largest purposes

Purpose	Applications	Funded Amount	Bad Loan %
Debt consolidation	18,214	\$232.46M	14.55%
credit card	4,998	\$58.89M	10.16%
other	3,824	\$31.16M	15.35%
home improvement	2,876	\$33.35M	11.37%
major purchase	2,110	\$17.25M	9.76%
small business	1,776	\$24.12M	25.62%
car	1,497	\$10.22M	10.35%

High-volume / higher-risk purpose screening

Using the same **2% portfolio-volume threshold** and above-portfolio Bad Loan rate:

- **Small business:** 1,776 applications, **25.62%** Bad Loan rate
- **Other:** 3,824 applications, **15.35%** Bad Loan rate
- **Debt consolidation:** 18,214 applications, **14.55%** Bad Loan rate

Insight

Debt consolidation dominates both application volume and funded exposure. Small business shows a substantially higher observed Bad Loan rate while still meeting the project's meaningful-volume screening rule.

8.9 Home Ownership Analysis

Home Ownership	Applications	Funded Amount	Bad Loan %	Avg DTI
RENT	18,439	\$185.77M	14.57%	13.50%
MORTGAGE	17,198	\$219.33M	12.97%	13.17%
OWN	2,838	\$29.60M	13.99%	13.24%
OTHER	98	\$1.04M	18.37%	11.25%
NONE	3	\$16.8K	0.00%	11.75%

Insight

- RENT is the largest category by application volume.
- MORTGAGE has the largest funded exposure.
- The major categories have relatively similar average DTI values.
- Very small groups such as **NONE** and **OTHER** should not be over-interpreted because their application counts are small.

7. Additional Risk Insights

9.1 Grade vs Bad Loan %

Grade	Applications	Avg Interest Rate	Avg DTI	Bad Loan %
A	9,689	7.35%	12.04%	5.70%
B	11,674	11.03%	13.43%	11.50%
C	7,904	13.55%	13.92%	16.02%
D	5,182	15.71%	13.98%	20.69%
E	2,786	17.71%	14.10%	24.80%
F	1,028	19.74%	14.17%	30.25%
G	313	21.40%	14.06%	31.31%

Insight

Observed Bad Loan rates increase across the grade spectrum in this dataset, while average interest rates also increase from Grade A to Grade G.

This is an observed segmentation pattern, not a causal claim that grade itself causes loan performance.

9.2 Grade vs Interest Rate

Average interest rate increases across the grade categories:

- Grade A: **7.35%**
- Grade B: **11.03%**
- Grade C: **13.55%**
- Grade D: **15.71%**
- Grade E: **17.71%**
- Grade F: **19.74%**
- Grade G: **21.40%**

Insight

Grade provides a clear view of observed pricing differences across the credit-grade spectrum.

9.3 Sub-Grade Risk Drill-Down

The sub-grade analysis provides more detailed segmentation within the broad grades.

Among the ten highest observed Bad Loan-rate sub-grades shown in the project visualization:

- **F5** has the highest observed Bad Loan rate at approximately **45.9%**.
- **G3** is approximately **39.5%**.
- **G2** is approximately **35.9%**.
- Several F and G sub-grades are clustered around the upper end of the observed risk range.

Insight

Sub-grade provides finer risk segmentation, but lower-volume groups should always be interpreted with their application counts.

9.4 Verification Status vs Bad Loan %

Verification Status	Applications	Funded Amount	Bad Loan %
Not Verified	16,464	\$139.70M	12.24%
Source Verified	9,777	\$99.10M	14.14%
Verified	12,335	\$196.96M	15.70%

Insight

Observed Bad Loan rates differ across verification categories. In this dataset, the **Verified** group has the highest observed Bad Loan rate among the three categories. This should be treated as a portfolio-

9.5 DTI vs Loan Status

Loan Status	Applications	Avg DTI	Median DTI	Std. Dev.
Fully Paid	32,145	13.17%	13.21%	6.67%
Charged Off	5,333	14.00%	14.32%	6.56%
Current	1,098	14.72%	15.01%	6.74%

Insight

Average DTI differs across loan-status groups. The boxplot also shows distributional differences, which provides more context than relying on average DTI alone.

9.6 Term × Loan Status — Average Installment

Key observed average installments:

- 36-month Charged Off: **\$316.79**
- 36-month Fully Paid: **\$313.95**
- 60-month Charged Off: **\$372.17**
- 60-month Current: **\$398.92**
- 60-month Fully Paid: **\$351.72**

Insight

Installment values differ across term and loan-status groups. Mean and median installments are both useful because installment values are not necessarily evenly distributed.

8. Supporting EDA

10.1 Annual Income vs Loan Amount

Purpose

Understand whether borrower income and loan size move together.

Observed pattern

- Loan amounts are concentrated mainly in lower-to-middle income ranges.
- Higher-income observations extend across a wide range of loan amounts.
- The plot shows substantial variation rather than a simple one-to-one relationship.
- The visualization was sampled to keep the scatter plot readable.

Business takeaway

Income and loan amount should be considered together when examining borrower capacity, rather than assuming that higher income automatically corresponds to a specific loan size.

10.2 Loan Amount Distribution

Statistic	Value
Count	38,576
Mean	\$11,296.07
Median	\$10,000
Minimum	\$500
25th percentile	\$5,500
75th percentile	\$15,000
99th percentile	\$35,000
Maximum	\$35,000

Insight

- The median loan amount is **\$10,000**.
- Half of the observed loans fall between 5, 500*and* **15,000**.
- The dataset is concentrated around common loan sizes while still covering a wider range up to \$35,000.

9. KPI Summary

Portfolio KPIs

KPI	Value
Total Applications	38,576
Total Funded	\$435.76M
Total Amount Received*	\$473.07M
Average Interest Rate	12.05%
Average DTI	13.33%

MTD KPIs

KPI	Value
Reporting Date	12-Dec-2021
MTD Applications	4,314
MTD Funded	\$53.98M
MTD Amount Received*	\$58.07M

Good vs Bad KPIs

KPI	Good	Bad
Applications	33,243	5,333
Share	86.18%	13.82%
Funded Amount	\$370.22M	\$65.53M
Loan-Level Payment	\$435.79M	\$37.28M

Important: The payment figures above come from the cumulative loan-level `total_payment` field. They are not transaction-level monthly cash collections.

10. Key Insights

Portfolio Overview

- **38,576** loan applications represent approximately **\$435.76M** in funded principal.
- Average interest rate is **12.05%**.
- Average DTI is **13.33%**.
- Recorded cumulative loan-level payment is approximately **\$473.07M**.

Loan Performance

- Under the project's analytical classification, **86.18%** of loans are Good and **13.82%** are Bad.
- Good Loans account for approximately **\$370.22M** in funded principal.
- Bad Loans account for approximately **\$65.53M**.

Origination Trend

- Origination activity generally increased through the observed 2021 period.
- Applications rose from **2,332 in January** to **4,314 in the observed December period**.
- Funded amount increased from **\$25.03M** to **\$53.98M**.
- December is partial through **12-Dec-2021**.

Geographic Concentration

- California has the largest observed application volume with **6,894 applications**.
- Geographic concentration and observed Bad Loan performance should be reviewed together.

Loan Characteristics

- 36-month loans dominate volume with **28,237 applications**.
- Debt consolidation is the largest purpose with **18,214 applications** and **\$232.46M** funded.
- 10+ years employment is the largest employment segment with **8,870 applications**.
- RENT is the largest home-ownership category by volume.
- MORTGAGE has the largest funded exposure.

Risk Interpretation

- Bad Loan rates vary across states, terms, purposes, employment groups, home ownership categories, grades, sub-grades and verification status.
 - Grade A has an observed Bad Loan rate of **5.70%**, while Grade G is **31.31%**.
 - 60-month loans show an observed Bad Loan rate of **22.34%**, compared with **10.71%** for 36-month loans.
 - Small-volume categories should be interpreted carefully alongside application counts and funded exposure.
 - The analysis identifies observed differences; it does not establish causation.
-

11. Business Interpretation

The analysis connects portfolio size, lending activity, borrower characteristics and observed loan performance into a single business story.

Portfolio Scale & Financial Position

- The portfolio represents a substantial observed lending base of **38,576 applications** and **\$435.76M** in funded principal.
- Average interest rate of **12.05%** provides a portfolio-level view of pricing.
- Average DTI of **13.33%** provides a broad borrower debt-burden view.
- Recorded payment should be interpreted as cumulative loan-level payment rather than a monthly collection ledger.

Loan Performance

- The majority of observed loans fall into the project's Good Loan category.
- Bad Loans account for **5,333 applications** and **\$65.53M** of funded principal.
- Application count and funded-dollar exposure provide complementary views of portfolio performance.

Origination Trends

- Lending activity generally increased across the observed year.
- Both application volume and funded amount moved upward from the early months toward the later observed months.
- December should be treated as a partial period because the dataset ends on 12-Dec-2021.

Portfolio Concentration

- Lending activity is highly concentrated in a few major segments: California, 36-month loans, debt consolidation, 10+ years of employment, RENT and MORTGAGE.
- High-volume segments are not automatically high-risk segments; volume and observed Bad Loan performance need to be reviewed together.

Risk & Pricing Insights

- Grade-level observed Bad Loan rates increase across the grade spectrum, while average interest rates also increase.
 - 60-month loans show higher observed Bad Loan rates and higher average interest rates than 36-month loans.
 - Purpose-level risk varies materially, with small business showing a particularly high observed Bad Loan rate among meaningful-volume screened purposes.
 - Verification categories also show different observed Bad Loan rates.
 - DTI and installment distributions provide additional context beyond portfolio averages.
-

12. Business Recommendations

The recommendations are framed as **monitoring, review and investigation actions**, not as causal conclusions.

1. **Monitor risk concentration** in meaningful-volume segments where observed Bad Loan rates are above the portfolio rate.
 2. **Track geographic exposure** using application volume, funded amount and observed Bad Loan performance together.
 3. **Review term-level behavior** using application mix, funded exposure, interest rate, Bad Loan rate and installment metrics together.
 4. **Monitor high-volume loan purposes** alongside funded exposure and observed Bad Loan rates.
 5. **Track borrower affordability patterns** using DTI, annual income, loan amount and installment together.
 6. **Review grade and sub-grade patterns** using pricing and observed loan performance.
 7. **Maintain data-quality controls** for missing fields, duplicate identifiers, dates and numeric business rules.
 8. **Use transaction-level payment data in a future phase** to support true monthly cash-collection and MTD collection analysis.
-

13. Conclusion

This project demonstrates an end-to-end Data Analyst workflow using Python, starting from business understanding and data-quality assessment through analysis, visualization, business interpretation, and recommendations.

Analytical Workflow

Business Understanding → Data Loading & Inspection → Data Quality Assessment → Data Cleaning → Feature Engineering → Core BRD & Business Analysis → Additional Risk Insights → Supporting EDA → KPI Summary → Key Insights → Business Interpretation → Business Recommendations

Key Portfolio Findings

- **38,576** loan applications represent approximately **\$435.76M** in funded principal.
- Average interest rate is **12.05%** and average DTI is **13.33%**.

- Under the project's analytical classification, **86.18%** of loans are Good Loans and **13.82%** are Bad Loans.
- Loan origination activity generally increased across the observed 2021 period, with December observed only through **12-Dec-2021**.
- Lending activity is concentrated across major state, term, employment, purpose, and home-ownership segments.
- Additional risk analysis provides views into **grade, sub-grade, verification status, DTI, and installment behavior**.

Skills Demonstrated

- **Python / Pandas**
- **Data Cleaning**
- **Feature Engineering**
- **Data Analysis**
- **Data Visualization**
- **Business-Oriented Analytical Thinking**

Final Project Perspective

The project is structured around business questions rather than isolated Python techniques. Each major analysis connects a business question with the appropriate metric, supporting table or visualization, and a practical interpretation.

The objective is to demonstrate the ability to move from **raw loan data to understandable business insights** while keeping the analysis focused, reproducible, and explainable in a fresher-level Data Analyst interview.

14. Quick Interview Revision

Explain the project in 30 seconds

"I analyzed a bank loan portfolio of 38,576 loan records using Python, Pandas, NumPy, Matplotlib and Seaborn. I started with business questions and data-quality assessment, cleaned the dataset, engineered analytical features, calculated portfolio and MTD KPIs, analyzed Good vs Bad Loans, and then investigated risk patterns across states, terms, purposes, employment, home ownership, grades, verification status and DTI. I finished by converting the findings into business insights and monitoring recommendations."

Most important numbers to remember

Metric	Value
Applications	38,576
Funded Amount	\$435.76M
Loan-Level Payment	\$473.07M
Avg Interest Rate	12.05%
Avg DTI	13.33%

Metric	Value
Good Loan %	86.18%
Bad Loan %	13.82%
Latest Issue Date	12-Dec-2021
Latest observed month applications	4,314
Latest observed month funded	\$53.98M
Largest state by applications	CA — 6,894
Largest purpose	Debt consolidation — 18,214
Largest employment group	10+ years — 8,870
Largest home-ownership group	RENT — 18,439
36-month Bad Loan %	10.71%
60-month Bad Loan %	22.34%
Grade A Bad Loan %	5.70%
Grade G Bad Loan %	31.31%

Three points to explain carefully in an interview

1. **Good/Bad classification:** Fully Paid + Current = Good; Charged Off = Bad. This is the project's analytical mapping.
2. **MTD limitation:** December is only observed through 12-Dec-2021.
3. **Payment limitation:** `total_payment` is cumulative loan-level payment, not transaction-level monthly cash collection.